

OUTAGE STATE RECONCILIATION ENGINE (OSRE)

Recovered an undocumented production platform and restored the outage-state capability supporting contact-center call deflection

ROLE

Solutions Architect (production recovery, technical lead)

IMPACT

Production recovery, formal API contract, safer client integration, restored call-deflection capability

THE CHALLENGE

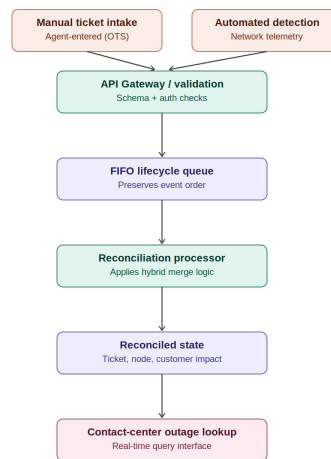
BUSINESS IMPACT

A production pipeline meant to auto-detect and close outage-related tickets—so contact-center callers with a known outage could be deflected—was misidentifying outages and leaving tickets open. Contact-center managers had already disabled reliance on it, absorbing the full unmitigated call volume it was built to reduce.

OPERATIONAL CONSTRAINT

The pipeline had zero documentation and its original owner had already left the company. No API contract existed with the client side, and the server side returned no response to requests at all—while executive leadership pushed for an accelerated fix.

ARCHITECTURE AT A GLANCE



DECISIONS THAT CHANGED THE OUTCOME

1 REVERSE-ENGINEER BEFORE CHANGING

With no documentation and no original author available, traced the deployed code directly to rebuild an understanding of the pipeline's logic before proposing any fix—under executive pressure to move fast.

2 DEFINE THE MISSING CONTRACT

Secured and documented an API contract with the client side that had never existed, after finding the server side wasn't returning any response to incoming requests.

3 RESOLVE HYBRID TICKET STATES

Fixed the logic merging manually reported and automatically detected outage events into a single reconciled state, correcting the root cause of tickets that weren't closing.

4 STABILIZE THE CLIENT ENDPOINT

Introduced an API Gateway custom domain so clients used a stable service URL instead of the provider-generated gateway endpoint, decoupling consumers from underlying deployment details.

MY CONTRIBUTION

INCIDENT OWNERSHIP

Took technical ownership of a broken, undocumented production deployment and diagnosed root causes directly from code under executive-level time pressure.

HANDS-ON DELIVERY

Corrected the outage-detection and ticket-reconciliation logic, defined the missing API contract, and restored synchronous server responses.

PRODUCTION READINESS

Introduced a stable custom-domain endpoint for API consumers and produced the architecture with IaC, API, observability, and handover documentation.

VERIFIED DELIVERY EVIDENCE

500K+

customers across the supported fixed-network footprint

18 / 18

documented internal acceptance scenarios passed

0 to defined

formal API contract where none previously existed

Internal ATP records 18/18 passed scenarios; additional independent testing was conducted by a separate unit. Production telemetry and customer data are excluded from this public portfolio.